Making Education Accessible and Reliable: AI Driven Facilitator Based Education Platform

Project Scope, Objectives and Literature Review

Moksh Manglani

Department of Software Engineering
The University of Auckland, Auckland, New Zealand

April 4, 2026



Supervisor: Gill Dobbie

Project Partner: May Thazin

Disclaimer *– All these conference papers have been submitted as partial fulfilment for the project requirement for the BE(Hons) degree. Although they have been assessed, no errors or factual information have been corrected or checked.*

Table of Contents

1. INTRODUCTION....3
2. PROJECT PLAN....3
    2.1 Project Overview....3
    2.2 Problem Context....3
    2.3 Problem Aim....4
    2.4 Objectives....4
    2.5 Scope....4
3. LITERATURE REVIEW....5
    3.1 Introduction to the Review....5
        3.1.1 Evaluation criteria....5
    3.2 Is There a Global Shortage of Quality Education?....5
    3.3 Current Modes of Education....6
        3.3.1 Formal Schooling....6
        3.3.2 Homeschooling and Self-Directed Learning....6
        3.3.3 Online and Self-Paced Learning....7
        3.3.4 Synchronous and Blended Learning....7
        3.3.5 Learning Management Systems and Educational Platforms....8
        3.3.6 Alternative and Community-Based Delivery Models....8
    3.5 Evidence for Structured and Facilitator-Supported Learning....9
    3.6 Identified Market and Research Gap....9
        3.6.1 Existing Models Perform Strongly on Some Criteria but Not All....9
        3.6.2 The Core Gap Is Between Scalable Access and Reliable Delivery....9
        3.6.3 Implications for the Proposed Model....10
    3.7 Conclusion....10
4. PROJECT SCHEDULE GANTT CHART-....11
    Figure 4.1: Project Timeline & Gantt Chart....11
REFERENCES....12

1. INTRODUCTION

Access to quality education remains one of the most pressing global challenges in social and economic development. Although education is widely recognised as a fundamental human right, many communities continue to experience limited access to reliable, affordable, and effective learning opportunities. This issue is particularly severe in rural, remote, and underserved settings, where shortages of qualified teachers, inconsistent educational delivery, and infrastructural constraints reduce both access and learning quality.

This project responds to that challenge through the design of a facilitator-based education platform intended to make educational delivery more accessible, affordable, and consistent. The central idea is to separate educational expertise from classroom delivery: lessons are designed by experts, then delivered by non-expert facilitators through a structured, step-by-step platform interface. In this way, the system seeks to lower the barriers to starting and operating educational programmes without requiring every classroom to depend on a fully qualified teacher. The project also incorporates AI-supported resource generation within the platform, although the primary focus of this document is on the broader educational delivery model rather than AI in isolation.

This report is structured in two distinct parts. The first presents the project plan, including the project overview, problem context, aims, scope, objectives, and methodology. The second presents the literature review, which examines how well existing modes of education address the global shortage of quality education and identifies the market and research gap that motivates the proposed facilitator-based model.

2. PROJECT PLAN

2.1 Project Overview

This project aims to design and develop a proof-of-concept facilitator-based education platform that supports the delivery of structured lessons by non-expert facilitators. The platform is intended to enable education providers to organise expert-designed lessons and distribute them through guided classroom delivery, thereby improving consistency and reducing dependence on qualified teacher availability at the point of instruction.

The concept behind the project is that high-quality teaching expertise does not necessarily need to reside in every classroom if it can instead be embedded into the lesson design, instructional flow, and support systems used during delivery. Under this model, experts design the lessons, materials, and activities centrally, while facilitators use the platform to follow a guided sequence during class. This provides a way to preserve structure and consistency while widening the pool of people who can participate in educational delivery.

The project includes the design of a facilitator dashboard for lesson execution, an administrative interface for creating and organising lesson content, and a platform structure capable of supporting multiple organisations. The intended outcome is a functional proof of concept that demonstrates the feasibility of structured facilitator-led lesson delivery in a platform-based environment..

2.2 Problem Context

The project is motivated by several interconnected educational challenges. First, there is a severe shortage of qualified teachers globally, particularly in low-income, rural, and remote areas. This shortage limits the expansion of formal education systems and contributes to inconsistent instructional quality. Second, student learning outcomes remain highly uneven, even among students who are enrolled in school, indicating that access to schooling alone does not ensure access to quality education. Third, establishing and operating a traditional educational institution is often expensive and administratively complex, making it difficult for underserved communities to create sustainable learning opportunities.

Existing educational models tend to address only part of this problem. Formal schooling offers structure and legitimacy but depends heavily on qualified teachers and institutional infrastructure. Online and self-paced learning can scale more easily, but often lacks the human guidance, accountability, and structure needed by many learners, especially younger children. Learning management systems support administration but do not solve the challenge of how to guide lesson delivery in classrooms where expert teachers are unavailable.

These conditions create a need for alternative delivery models that can improve accessibility, consistency, and affordability without depending entirely on qualified teacher availability. The proposed facilitator-based model is designed in response to this problem context.

2.3 Problem Aim

The aim of this project is to design and develop a proof-of-concept facilitator-based education platform that enables non-expert facilitators to deliver expert-designed lessons through structured digital guidance, thereby supporting more accessible, affordable, and consistent educational delivery.

2.4 Objectives

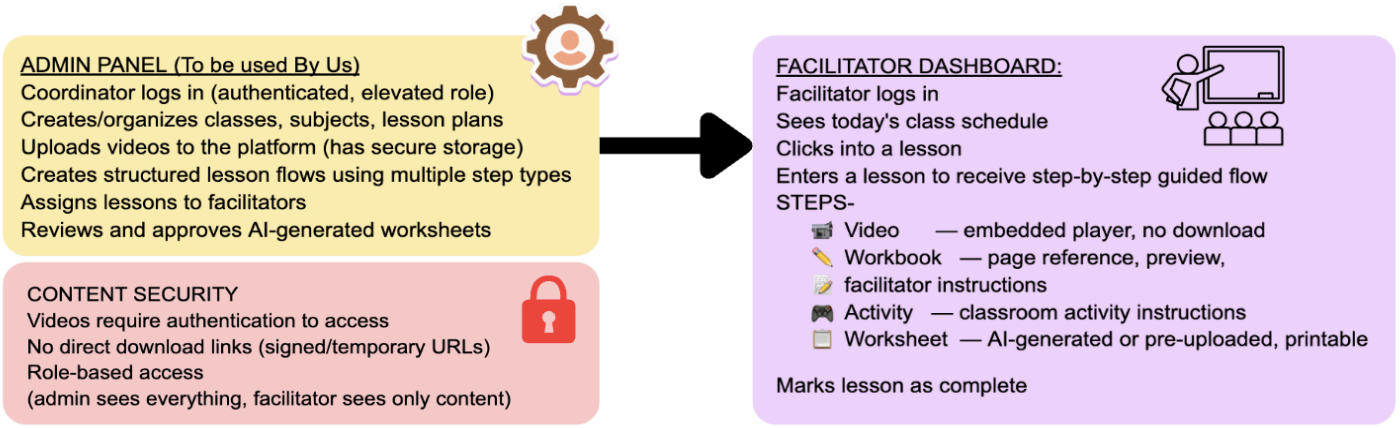
The objectives of the project are to:

- investigate the educational problem space surrounding access, quality, and teacher dependency
- identify the limitations of current modes of educational delivery
- design a platform model that separates lesson design from classroom delivery
- develop a facilitator dashboard that supports step-by-step lesson execution
- develop an administrative interface for creating and managing structured lesson flows
- incorporate platform features that support resource generation and content management
- demonstrate the feasibility of the model through a working proof of concept
- evaluate the design concept against the needs and gaps identified in the literature

2.5 Scope

The scope of this project is limited to the design and implementation of a proof-of-concept platform for facilitator-based lesson delivery. The project focuses on the structure and functionality required to support guided classroom instruction rather than on the full deployment of a complete educational business or institution.

System mockup-



To understand or scope deeply, it is also crucial to understand what is out of our scope for this project-

- large-scale field deployment
- long-term educational impact evaluation
- complete payment, subscription, or marketplace systems
- full regulatory compliance workflows for operating schools
- comprehensive student assessment systems beyond basic demonstration features

3. LITERATURE REVIEW

3.1 Introduction to the Review

This literature review examines how well existing modes of education address the global shortage of quality education. Rather than treating education simply as school access or enrollment, this review is concerned with whether current educational models are capable of delivering meaningful, reliable, and scalable learning. This distinction is important because many learners remain excluded not only from schooling itself, but from quality educational experiences that produce strong and consistent outcomes (UNESCO, 2020; World Bank, 2025).

To evaluate current modes of education, this review applies six analytical criteria drawn from recurring themes in the literature: **personalisation, decoupled delivery, structured guidance, accessibility, adaptability, and engagement**. Together, these criteria provide a framework for assessing not only whether education is available, but whether it can be delivered in a way that is responsive to learner needs, less dependent on scarce professional expertise, and effective across varied educational contexts.

Using these criteria, the review first establishes whether there is a global shortage of quality education. It then examines major modes of educational delivery and evaluates their strengths and limitations. Finally, it identifies the market and research gap that supports the case for a facilitator-based model.

3.1.1 Evaluation criteria

Personalisation is an educational model's ability to respond to differences in learner ability, pace, and need. Personalised learning has been shown to improve engagement, achievement, and motivation by aligning instruction with individual learners (Hooshyar et al., 2024).

Decoupled delivery refers to the extent to which educational expertise can be separated from direct classroom instruction. This metric captures how strongly the delivery of education depends on trained professionals. It is especially relevant in contexts facing teacher shortages, where reliance on qualified educators can constrain the reach, scalability, and resilience of education systems (UNESCO, 2024; Veine et al., 2022).

Structured guidance is the provision of clear instructional sequences, step-by-step lesson flows, and defined support materials that reduce ambiguity in delivery. Structured pedagogy has been identified as one of the most effective approaches for improving learning outcomes, particularly in low-resource settings where consistency of delivery is difficult to maintain (Piper et al., 2018; Snilstveit et al., 2016; World Bank, 2018).

Accessibility is an educational model's ability to overcome geographic, economic, and institutional barriers and make quality education more widely available. This remains a central concern in global education, where inequity is closely tied to poverty, remoteness, and limited resources (UNESCO, 2020; World Bank, 2025).

Adaptability is the capacity of an educational model to be adjusted across different learners, subjects, curricula, and contexts. This is important because learning environments vary significantly across age groups, institutions, and cultural settings, and rigid models often struggle to transfer effectively across them (Achuthan, 2025; Lang, 2025).

Engagement is the degree to which an educational model can sustain learner attention, participation, and involvement in the learning process. High levels of engagement are associated with stronger learning outcomes, while low engagement remains a common limitation of self-paced and online learning systems (Garrison et al., 2000; Jordan, 2015; Means et al., 2010).

3.2 Is There a Global Shortage of Quality Education?

The literature clearly suggests that the world faces a shortage not merely of schooling, but of quality education. Education is widely recognised as a fundamental human right and is central to Sustainable Development Goal 4, which seeks to ensure inclusive and equitable quality education for all (The Global Goals, 2025). Despite this commitment, access to meaningful educational opportunities remains uneven across countries and communities, especially in contexts affected by poverty, underinvestment, conflict, and geographic isolation (UNESCO, 2020).

One of the clearest indicators of this shortage is the global teacher crisis. UNESCO (2024) estimated that 44 million additional primary and secondary teachers are needed worldwide by 2030 to achieve universal education goals. This shortage is particularly severe in low-income and rural settings, where schools frequently struggle to recruit and retain qualified staff. In such environments, delivery remains heavily dependent on a limited pool of trained

professionals, making education systems less accessible and less resilient. This has direct implications for decoupled delivery, since most current systems still rely heavily on qualified teachers at the point of instruction.

The scale of the quality problem is further reflected in learning outcomes. The World Bank (2025) estimates that 53% of children in low- and middle-income countries experience learning poverty, defined as being unable to read and understand a simple text by age 10. This suggests that school participation alone does not guarantee meaningful learning. Even where learners are enrolled, educational models may still fail in terms of structured guidance, engagement, and personalisation, resulting in poor foundational outcomes. Global learning data also remains incomplete, indicating that the full scale of the problem may be underestimated (World Bank, 2025a).

Geographic inequality further compounds this crisis. Rural and remote communities are disproportionately affected by limited teacher availability, poor infrastructure, and restricted access to educational resources. These barriers reduce accessibility and make it difficult to deliver consistent educational quality. Taken together, the literature supports the conclusion that the current global education crisis is not only about whether education exists, but whether it is accessible, engaging, adaptable, and reliably delivered across different contexts.

3.3 Current Modes of Education

3.3.1 Formal Schooling

Formal schooling remains the dominant and most institutionally recognised mode of education. It is typically organised around structured curricula, trained teachers, age-based progression, and regulated assessment. In terms of structured guidance, formal schooling performs strongly, as it usually provides sequenced instruction, classroom routines, and clear expectations for learning progression (UNESCO, 2015). It can also support meaningful engagement through direct teacher-student interaction, peer discussion, and social learning environments. Formal education additionally offers recognised pathways for certification and progression, making it the default model for quality assurance in many systems.

Formal schooling can also support a degree of personalisation, particularly when teachers are able to differentiate instruction in response to student needs. However, this varies significantly depending on class size, teacher workload, and available resources. In large or under-resourced classrooms, personalisation is often limited by the practical demands placed on individual teachers.

The model performs less strongly in terms of accessibility and decoupled delivery. Formal schooling depends heavily on infrastructure, staffing, and institutional coordination, making it difficult to establish and sustain in underserved regions. Its effectiveness is also strongly tied to the availability of qualified teachers, meaning that expertise is not separated from delivery. This creates significant bottlenecks in areas affected by teacher shortages (UNESCO, 2024). In terms of adaptability, formal schooling can also be slow to respond to different local needs, curricula, or cultural contexts because of its regulatory and administrative rigidity. As a result, while formal schooling remains strong in structure and legitimacy, it is often difficult to scale equitably across diverse settings.

3.3.2 Homeschooling and Self-Directed Learning

Homeschooling and self-directed learning provide greater flexibility than formal schooling and may allow learners to work at their own pace. In principle, these models can support strong personalisation, as content and pacing can be adapted more directly to the learner's needs and interests. They may also offer some adaptability, particularly where families or learners can tailor the curriculum or learning environment to suit specific goals and contexts.

However, the effectiveness of these models depends heavily on the learner's capacity for self-regulation and the quality of support available in the home environment. Research on self-directed learning in online and digital settings suggests that learners need strong planning, monitoring, and self-management skills in order to benefit from independent learning systems (Chen, 2009; Kim et al., 2014). For younger learners in particular, this often results in weak structured guidance and inconsistent engagement unless a highly capable adult is present to provide support.

Homeschooling and self-directed learning also perform unevenly in terms of accessibility, since successful implementation often depends on parental time, educational confidence, and access to quality resources. In this sense, they may widen inequities rather than reduce them. They also do little to address decoupled delivery,

because educational provision still depends heavily on the capability of the adult directly supporting the learner. Although these models can be effective in particular contexts, the literature does not support them as a broadly scalable solution to the global shortage of quality education.

3.3.3 Online and Self-Paced Learning

Online and self-paced learning platforms such as Khan Academy, Coursera, and Udemy have greatly expanded access to educational content. In terms of accessibility, these models perform strongly, as they reduce geographic barriers and make learning materials available at low cost and large scale. They can also support some degree of personalisation, particularly where systems adjust pacing, recommend content, or allow learners to revisit material according to their needs.

However, the literature consistently shows that self-paced online learning struggles in terms of engagement and structured guidance. A MOOC (Massive Open Online Course) is a free, web-based distance learning program designed for a large number of geographically dispersed participants. Jordan (2015) found that MOOC completion rates were often below 15%, suggesting that learners frequently disengage when left to navigate learning independently. Kizilcec et al. (2013) similarly found that while many learners initially enrol in online courses, sustained participation is limited. These outcomes indicate that scalability and access do not necessarily translate into effective learning.

Self-paced systems also perform weakly in terms of decoupled delivery, though in a different way from formal schooling. Rather than reducing dependence on teachers by replacing them with structured facilitation, they often shift the burden of learning onto the student. This makes them especially problematic for younger learners, who typically need stronger instructional support and supervision. Their adaptability can vary: some platforms allow flexible use across topics and learner groups, while others are limited to specific subjects or standardised content structures. Overall, self-paced online education performs well in accessibility but weakly in engagement, guided delivery, and suitability for learners who require support.

3.3.4 Synchronous and Blended Learning

A synchronous learning model is an educational approach where instructors and students interact in real-time, engaging simultaneously from the same or different locations. It mirrors traditional classroom interaction through live video conferencing, instant messaging, or in-person sessions, allowing for immediate feedback, collaboration, and structured scheduling.

Synchronous online learning seeks to preserve some of the interactive qualities of classroom teaching by enabling live instruction through digital platforms. In terms of engagement, this model generally performs more strongly than self-paced learning because learners can interact with instructors and peers in real time. The Community of Inquiry framework emphasises the role of teaching presence, social presence, and interaction in meaningful online learning (Garrison et al., 2000), which helps explain why synchronous models often feel more educationally rich than purely asynchronous ones.

Blended learning extends this further by combining digital content with in-person or facilitated support. This model can perform relatively well across several criteria. It may improve engagement by combining flexibility with human interaction, and it can support more effective structured guidance when digital content is embedded within planned teaching sequences. Means et al. (2010) found that blended learning environments often outperform purely online or purely face-to-face formats, suggesting that combining modalities can produce stronger outcomes.

However, both synchronous and blended models remain limited in terms of decoupled delivery. They still depend heavily on skilled teachers or facilitators being present in real time or in person, which constrains scalability in contexts where qualified staff are scarce. Their accessibility is also affected by infrastructure requirements such as reliable internet, scheduling coordination, and device access. In addition, while these models can be adaptable, their implementation often requires substantial planning and institutional support. As a result, they offer stronger engagement and structure than self-paced systems, but remain difficult to scale equitably across underserved settings.

3.3.5 Learning Management Systems and Educational Platforms

Learning management systems (LMSs) such as Moodle, Canvas, Blackboard, and Google Classroom provide important infrastructure for organising educational content, assessment, and communication. These systems can support accessibility by making resources available digitally and can offer a degree of adaptability, as content can be rearranged, reused, or modified across subjects and classes. They may also support limited personalisation where teachers use them to distribute differentiated materials or assignments.

Despite this, LMSs perform relatively weakly in structured guidance when considered as stand-alone educational models. Coates et al. (2005) found that LMSs often function more as administrative repositories than active pedagogical systems. Their effectiveness depends largely on how teachers design lessons and manage delivery around them. In other words, LMSs support the organisation of education rather than the act of teaching itself.

They also do not perform strongly in decoupled delivery, since they assume the presence of teacher expertise to interpret content, lead activities, and sustain classroom engagement. In terms of engagement, LMSs can support communication and interaction, but this generally depends on the pedagogical design built around the platform rather than the platform alone. Therefore, while LMSs are useful educational tools, they do not by themselves resolve the central challenges of quality, consistency, and teacher dependency.

3.3.6 Alternative and Community-Based Delivery Models

Alternative and community-based education models, including para-teacher programmes, volunteer-led instruction, and non-formal learning initiatives, are particularly relevant to this review because they begin to challenge the assumption that all effective education must be delivered by fully qualified teachers. These models often perform better in terms of decoupled delivery, as they demonstrate that educational provision can, under certain conditions, be delivered by community members or minimally trained facilitators.

Initiatives such as Pratham's Teaching at the Right Level and BRAC's community-based models suggest that non-traditional delivery can be effective when it is supported by strong programme structure and targeted pedagogical design. In this sense, they also perform relatively well in structured guidance, particularly where content and routines are centrally defined. Their localised nature may further support accessibility, especially in underserved communities where formal schooling is weak or absent.

However, these models vary in their adaptability and engagement depending on how they are designed and implemented. Some programmes are highly responsive to local needs, while others are more rigid or dependent on external support. In addition, many community-based models are programme-specific rather than implemented through reusable or scalable digital systems. This limits their transferability and makes it difficult to generalise them as broad educational solutions. Nevertheless, they provide important evidence that educational expertise and delivery can be partly separated, particularly when strong support structures are in place.

3.4 Limitations of Existing Educational Models

Across the literature, a consistent pattern emerges: existing educational models tend to perform strongly on some of the evaluation criteria but poorly on others. Formal schooling is strong in structured guidance and often in engagement, but weak in decoupled delivery and often limited in accessibility where teacher supply and infrastructure are constrained. Homeschooling and self-directed learning can support personalisation and some adaptability, but often lack consistent structured guidance and are difficult to scale equitably. Self-paced online learning performs well in accessibility and sometimes in personalisation, but remains weak in engagement and guided instructional support. Synchronous and blended learning improve engagement and structure, yet still depend heavily on professional expertise and infrastructure. LMSs support administration and flexibility but do not themselves solve the problem of educational delivery.

The central limitation, therefore, is not that current models are ineffective in all respects, but that they do not balance the full set of needs implied by the global education crisis. Models that are highly accessible are often weak in structure and engagement, while models that are strong in guidance and interaction are often difficult to scale or too dependent on qualified teachers. This is especially significant in low-resource settings, where the challenge is

not only to provide education, but to do so in ways that are affordable, resilient, and less vulnerable to professional shortages.

Another key issue is that current systems rarely combine decoupled delivery with strong structured guidance. Where educational expertise is separated from delivery, this often happens through self-paced content consumption rather than supported classroom facilitation. Conversely, where structure and engagement are strong, this usually depends on a qualified educator being present. This leaves a gap between scalable educational access and reliable, human-supported delivery.

3.5 Evidence for Structured and Facilitator-Supported Learning

The literature on structured pedagogy provides strong support for the value of clear and well-sequenced instructional design. Piper et al. (2018) found that structured pedagogical programmes in Kenya produced substantial gains in literacy, while Snilstveit et al. (2016) identified structured pedagogy as one of the most effective categories of intervention in low- and middle-income countries. The World Bank (2018) similarly recommended structured lesson plans and teacher support as high-impact strategies for improving foundational learning. These findings suggest that structured guidance is not a minor design feature, but a core contributor to educational effectiveness.

This is particularly relevant when considering decoupled delivery. If lesson content, teaching steps, and support materials are clearly designed in advance, it becomes more plausible that classroom delivery can be supported by people who are not fully qualified teachers. Research on facilitators and learning assistants supports this possibility. Veine et al. (2022) found that learning assistants can contribute meaningfully as facilitators when they are supported with training and structured responsibilities. Daga-As and Guntalidad (2025) similarly showed that alternative learning system facilitators can produce meaningful educational impact despite limited formal training, provided appropriate support systems are in place.

Facilitator-supported learning may also improve engagement, particularly in comparison with self-paced digital learning. Human presence in the classroom can support discipline, encouragement, and participation, even where the facilitator is not the original designer of the lesson. In addition, facilitator-based models may improve accessibility by widening the pool of individuals capable of delivering education, especially in communities where fully qualified teachers are scarce. This does not eliminate the need for expertise; rather, it changes where that expertise is located. Instead of being concentrated entirely in the classroom teacher, expertise can be embedded in lesson design, instructional sequences, and central content creation.

This body of literature does not prove that all facilitator-based models are equally effective, nor does it suggest that expertise becomes unnecessary. However, it does provide a theoretical and empirical basis for exploring models in which educational design and educational delivery are more clearly separated.

3.6 Identified Market and Research Gap

3.6.1 Existing Models Perform Strongly on Some Criteria but Not All

When evaluated against the six criteria used in this review, existing modes of education reveal a fragmented landscape. Formal schooling offers strong structured guidance and often strong engagement, but is highly dependent on qualified teachers and difficult to extend into underserved settings. Self-paced online learning performs strongly in accessibility and in some cases personalisation, but performs poorly in engagement and structured guidance. LMSs support adaptability and resource organisation, yet do not provide guided delivery or reduce dependence on educators. Alternative community-based models begin to address decoupled delivery, but are often programme-specific and not implemented through scalable platform systems.

3.6.2 The Core Gap Is Between Scalable Access and Reliable Delivery

The literature suggests that current educational models tend to divide into two broad groups: those that provide quality through expert-led, resource-intensive delivery, and those that provide access through low-cost, content-based delivery. What remains underdeveloped is a model that combines accessibility, structured guidance, engagement, and decoupled delivery in a coherent and scalable form. In particular, there is limited evidence of mainstream educational platforms designed specifically to guide non-expert facilitators through live lesson delivery using expert-created instructional sequences.

3.6.3 Implications for the Proposed Model

This gap is significant because it points to the possibility of a different educational architecture. If lessons can be centrally designed with strong pedagogical structure, then classroom delivery may not need to rely entirely on expert teachers. A facilitator-based model may therefore offer a way to preserve human interaction and classroom engagement while improving accessibility and reducing professional bottlenecks. From this perspective, the value of the proposed model lies not simply in digitising education, but in reconfiguring the relationship between expertise, delivery, and access.

3.7 Conclusion

This review has shown that the global shortage of quality education is a substantial and multidimensional problem, reflected in widespread learning poverty, persistent teacher shortages, and major inequalities in educational access and quality. Existing modes of education respond to this challenge in different ways, but none fully satisfy the combined demands of personalisation, decoupled delivery, structured guidance, accessibility, adaptability, and engagement.

The literature therefore suggests a clear gap for a model that combines the strengths of structured pedagogical design with a more flexible and accessible delivery model. A facilitator-based approach, in which lessons are designed by experts and delivered through structured support by non-expert facilitators, appears well positioned to address this gap. As such, it provides a plausible response to the limitations identified across existing educational models and offers a strong conceptual basis for the present project.

4. PROJECT SCHEDULE GANTT CHART-

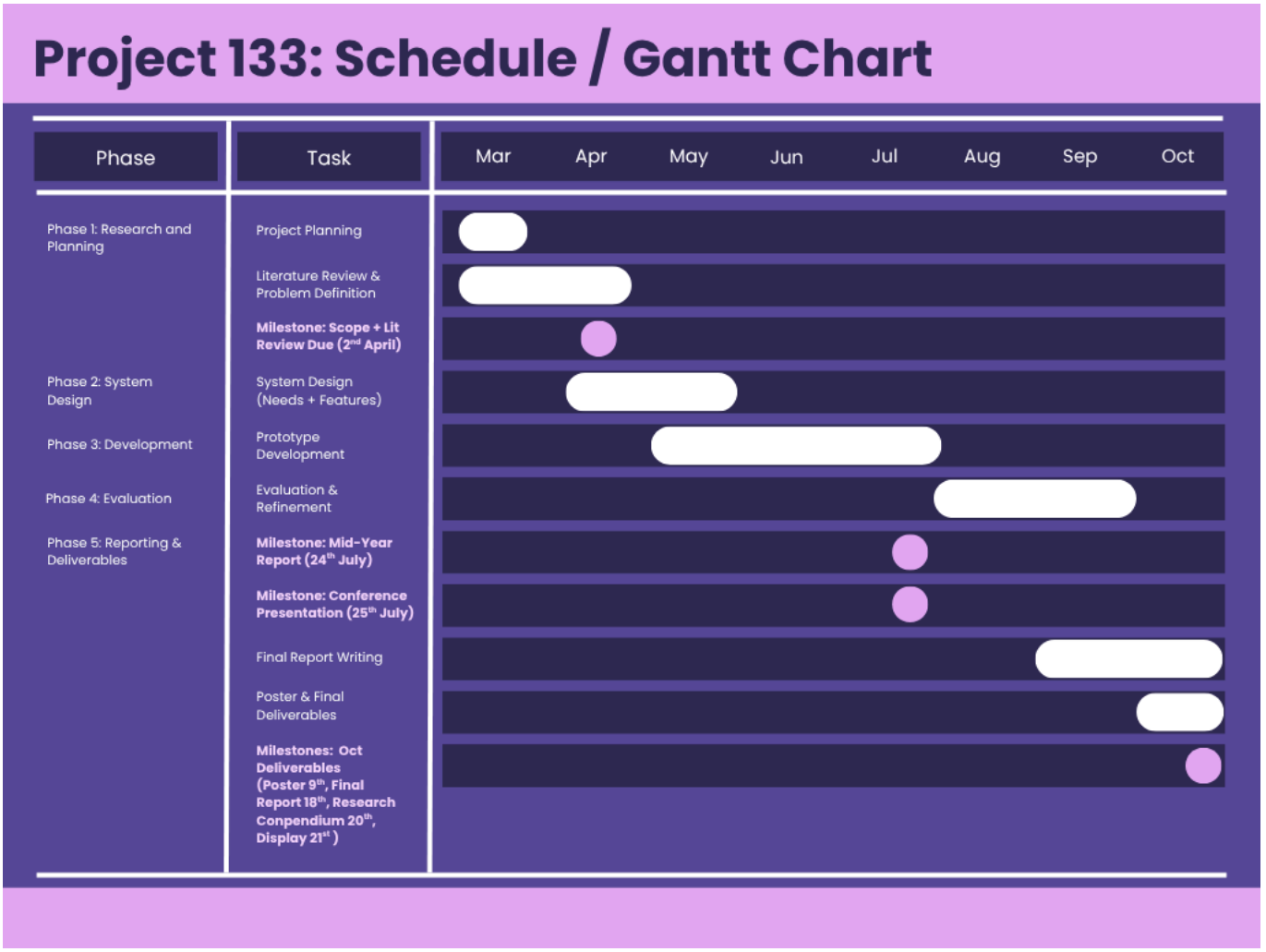


Figure 4.1: Project Timeline & Gantt Chart

The project timeline is structured across key phases, beginning with research and literature review, followed by system design, development, and evaluation. Major milestones, including the literature review submission, mid-year report, and final deliverables, are aligned with the course schedule. This phased approach ensures steady progress and allows sufficient time for refinement and completion of all project components.

REFERENCES

1) Achuthan, K. (2025). Artificial intelligence and learner autonomy: A meta-analysis of self-regulated and self-directed learning. Frontiers in Education, 10, Article 1738751. https://doi.org/10.3389/feduc.2025.1738751

2) Bailenson, J. N. (2021). Nonverbal overload: A theoretical argument for the causes of Zoom fatigue. Technology, Mind, and Behavior, 2(1). https://doi.org/10.1037/tmb0000030

3) Chen, C.-M. (2009). Personalized e-learning system with self-regulated learning assisted mechanisms for promoting learning performance. Expert Systems with Applications, 36(5), 8816–8829. https://doi.org/10.1016/j.eswa.2008.11.026

4) Coates, H., James, R., & Baldwin, G. (2005). A critical examination of the effects of learning management systems on university teaching and learning. Tertiary Education and Management, 11(1), 19–36. https://doi.org/10.1080/13583883.2005.9967137

5) Daga-As, C. P., & Guntalidad, J. A. (2025). Breaking boundaries: The real lives of alternative learning system facilitators. Journal of Educational Management and Strategy, 4(2), 133–142. https://doi.org/10.57255/jemast.v4i2.1518

6) Darling-Hammond, L. (2000). Teacher quality and student achievement. Education Policy Analysis Archives, 8(1), Article 1. https://doi.org/10.14507/epaa.v8n1.2000

7) Durkheim, E. (1956). Education and sociology. Free Press.

8) Garrison, D. R., Anderson, T., & Archer, W. (1999). Critical inquiry in a text-based environment: Computer conferencing in higher education. The Internet and Higher Education, 2(2–3), 87–105. https://doi.org/10.1016/S1096-7516(00)00016-6

9) Global Goals. (2025). Goal 4: Quality education. https://globalgoals.org/goals/4-quality-education/

10) Hooshyar, D., Weng, X., Sillat, P. J., Tammets, K., Wang, M., & Hämäläinen, R. (2024). The effectiveness of personalized technology-enhanced learning in higher education: A meta-analysis with association rule mining. Computers & Education, 223, Article 105169. https://doi.org/10.1016/j.compedu.2024.105169

11) Huang, R., Yin, Y., Zhou, N., & Lang, F. (2026). Artificial intelligence in K–12 education: An umbrella review. Computers and Education Artificial Intelligence, 10, Article 100519. https://doi.org/10.1016/j.caeai.2025.100519

12) Jordan, K. (2015). Massive open online course completion rates revisited: Assessment, length and attrition. The International Review of Research in Open and Distributed Learning, 16(3), 341–358. https://doi.org/10.19173/irrodl.v16i3.2112

13) Kim, R., Olfman, L., Ryan, T., & Eryilmaz, E. (2014). Leveraging a personalized system to improve self-directed learning in online educational environments. Computers & Education, 70, 150–160. https://doi.org/10.1016/j.compedu.2013.08.006

14) Kizilcec, R. F., Piech, C., & Schneider, E. (2013). Deconstructing disengagement: Analyzing learner subpopulations in massive open online courses. In Proceedings of the third international conference on learning analytics and knowledge (pp. 170–179). ACM. https://doi.org/10.1145/2460296.2460330

15) Means, B., Toyama, Y., Murphy, R., Bakia, M., & Jones, K. (2010). Evaluation of evidence-based practices in online learning: A meta-analysis and review of online learning studies. U.S. Department of Education.

16) National Center for Education Statistics. (2003). National Household Education Surveys Program of 2001 [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/ICPSR03198.v1

17) Piper, B., DeStefano, J., Kinyanjui, E. M., & Ong'ele, S. (2018). Scaling up successfully: Lessons from

Kenya's Tusome national literacy program. Journal of Educational Change, 19(3), 293–321. https://doi.org/10.1007/s10833-018-9325-4

18) Snilstveit, B., Stevenson, J., Menon, R., Phillips, D., Gallagher, E., Geleen, M., Jobse, H., Schmidt, T., & Jimenez, E. (2016). The impact of education programmes on learning and school participation in low- and middle-income countries: A systematic review summary report. International Initiative for Impact Evaluation (3ie).

19) United Nations Educational, Scientific and Cultural Organization. (2015). Rethinking education: Towards a global common good? UNESCO.

20) United Nations Educational, Scientific and Cultural Organization. (2020). Global education monitoring report 2020: Inclusion and education: All means all. UNESCO. https://unesdoc.unesco.org/ark:/48223/pf0000373718

21) United Nations Educational, Scientific and Cultural Organization. (2024). Global report on teachers: Addressing teacher shortages and transforming the profession. https://www.unesco.org/en/articles/global-report-teachers-addressing-teacher-shortages-and-transforming-profession

22) Veine, S., Anderson, M. K., Skancke, L. B., & Wallin, P. (2022). Educating learning assistants as facilitators: Design challenges and experiences of practice. Journal of Experiential Education, 46(4), 491–512. https://doi.org/10.1177/10538259221147010

23) World Bank. (2018). World development report 2018: Learning to realize education's promise. World Bank.

24) World Bank. (2025a). Learning poverty data. https://data360.worldbank.org/en/dataset/WB_LPGD

25) World Bank. (2025b). Learning poverty data and reporting gaps. https://openknowledge.worldbank.org/entities/publication/56452932-59c1-4a10-9216-1bea97e2de8a